Institute/ School Name	School of Engineering and Technology		
Department Name	Department of Computer Science & Engineering		
Program Name	Bachelor of Engineering (Computer Science & Engineering): B.E (CSE)		
Course Code	25IP003	Course Name	Integrated Project-III
L-T-P (Per Week)	0-0-4	Course Credits	02
Academic Year	2025-26	Semester/Batch	6th /2023-27
Course Coordinator	Dr. Nagesh Kumar		

1. Course Outline:

The purpose of the Integrated Project is to help students bring their knowledge, vision, and ethical considerations to life. This approach enhances the quality of the final product and promotes unbiased team dynamics, ensuring equitable participation irrespective of gender. By fostering an inclusive and collaborative environment, the project work equips students with the necessary skill sets to solve engineering-related problems while upholding principles of professional ethics and teamwork in a diverse setting.

2. Programme Outcomes (POs):

At the end of the programme, students will be able to achieve knowledge about the following:	
PO 1	**Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO 2	**Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	**Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO 4	**Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO 5	**Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO 6	**The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO 7	**Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO 8	**Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 9	**Individual and teamwork**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO 10	**Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO 11	**Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO 12	**Life-long learning**: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

3. Course Learning Outcomes (CLO):

After completing the course, the students will be able to:

CLO1: Understand use cases and formulate effective solutions by developing analytical skills.
CLO2: Demonstrate technical knowledge of the underlying technologies of the project, enhancing employability by mastering industry-relevant skills.
CLO3: Identify, analyze, and effectively solve practical and real-world problems.
CLO4: Design engineering solutions to complex problems using innovation and problem-solving skills.
CLO5: Apply engineering knowledge and ethical practices to solve industrial problems, improving employability in engineering and related fields.

4. CLO-PO Mapping Matrix:

Course Learning Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CLO1	M	M				L	L			L	M	L
CLO2		H	H	H		L	L				L	
CLO3		M		H		L						
CLO4	M	M		H	H	H		L	L		M	
CLO5	H	H						M				L

5. ERISE Grid Mapping:

Feature Enablement	Level (1-5, 5 being highest)
Entrepreneurship	2
Research/Innovation	3
Skills	5
Employability	3

6. Evaluation Scheme & Components:

Evaluation Component	Type of Component	No. of Assessments	Weightage of Component	Mode of Assessment (Offline/Online)
External Component	End Term Examination	01*	100%	Offline
Total		**100%**		

7. Revision (if any):

Academic Yearof Previous Version	-	Percentage of Revision	-
Topics:			

8. This Document is:

Designation	Name	Signature
Prepared by Course Coordinator	Dr. Nagesh Kumar	Nagesh
Verified by Assistant Dean	Dr. Ashutosh Kumar Dubey | Dr. Ravita Chahar	
Date	January 10, 2026	